

T6: Making the most of cis-regulatory annotations for genome-scale analysis

This full-day tutorial provides tools for the interpretation, retrieval and visualisation of currently available resources defining regulatory elements in multiple species and providing primary epigenomic data processed in a standardised way that makes it reusable and easily integrable with different bioinformatic workflows. The training combines conceptual framing, guided browser exploration, reproducible file-based workflows, and an integrative case study. The emphasis throughout is on distinguishing evidence layers: raw or processed signal, peak tracks, regulatory annotations, gene annotation and user-defined regions. Although these resources are publicly available and freely accessible, analysing and interpreting them remains challenging.

By the end of the tutorial, participants will be able to:

- Explore different resources offering raw and processed large-scale epigenomics data.
- Describe how regulatory elements in the genome are inferred from epigenomic assays such as ATAC-seq, DNase-seq, histone ChIP-seq, and TF ChIP-seq.
- Distinguish between major regulatory annotation frameworks and outline their underlying evidence (e.g. Ensembl Regulatory Build, ENCODE cCREs, FANTOM5, RefSeq Functional Elements, etc).
- Navigate and interpret regulatory annotations using Ensembl Regulation, the Ensembl Genome Browser, and SCREEN.
- Visualise regulatory elements and supporting evidence using genome browsers like IGV.
- Retrieve, parse, and analyse regulatory annotations in R and Python.
- Filter regulatory annotations by feature class, tissue, cell type, epigenome, assay, and activity context.
- Visualise regulatory evidence in IGV and interpret concordance and conflicts between regulatory annotation resources.
- Produce cautious, provenance-aware interpretation statements for non-coding variants or genomic regions.

Abstract

Cis-regulatory elements, such as promoters and enhancers, play a central role in gene regulation. Effective use of regulatory annotations is therefore essential for elucidating the regulatory control of specific genomic loci, identifying tissue-specific enhancers across the genome, and filtering and interpreting non-coding genetic variation. Large-scale community efforts, including ENCODE for human and mouse, and Functional Annotation of Animal Genomes (FAANG) projects for farmed animals, have generated thousands of epigenomic datasets, which are distributed across multiple repositories. Taking advantage of these datasets, ENCODE has recently expanded its registry of candidate cis-regulatory elements for human and mouse, while Ensembl has produced a comprehensive catalogue of regulatory features for ten species (human, mouse, three livestock, and five aquaculture species). Although these resources are publicly available and freely accessible, analysing and interpreting them remains challenging.

This tutorial emphasises the complementary roles of key resources and tools, highlighting best practices for accessing regulatory annotations, understanding their supporting evidence, and integrating them into reproducible genomic workflows. It provides a practical introduction to

regulatory genomics, focusing on how regulatory elements are derived from epigenomic data and how they can be explored across widely used community resources. Participants will learn how to navigate and compare regulatory annotations using the Ensembl Genome Browser (<https://www.ensembl.org/>), the new Ensembl Regulation subsite (<https://regulation.ensembl.org/>), and the ENCODE SCREEN portal (<https://screen.wenglab.org/>). The tutorial will also discuss experimentally validated and manually curated regulatory resources, highlighting how these complement large-scale inferred regulatory annotations. The tutorial further demonstrates how to manipulate and visualise regulatory data using R and Python, as well as genome browsers such as IGV.

Target audience and prerequisites

This tutorial is aimed at bioinformaticians, computational biologists, researchers and those working in functional genomics and gene regulation in biomedical or agricultural research.

Participants are expected to have:

- A basic understanding of genomics concepts (genes, genomic coordinates, genome assemblies).
- Familiarity with either R or Python.
- Basic command-line or notebook experience is helpful but not essential.

Tutorial Speakers

- Gabriela A. Merino, Ensembl Regulation, EMBL-EBI
- Garth Ilesley, Ensembl Regulation, EMBL-EBI
- Louise Paola Mirabueno, Ensembl Outreach, EMBL-EBI

Tutorial Format

The tutorial uses a progressive approach:

- Build conceptual understanding of regulatory elements and epigenomic evidence.
- Explore regulatory catalogues in Ensembl Regulation and SCREEN.
- Move from browser views to reusable annotation files and programmatic analysis.
- Visualise multiple evidence layers and annotation tracks in IGV.
- Integrate all methods in a guided non-coding variant interpretation case study.

The hands-on sections use provided datasets and notebooks so that participants can focus on interpretation rather than large downloads. Participants may choose the R path, the Python path, or compare both.

Required Tools and Materials

Instructions for installing required tools, libraries and packages will be sent in advance. Participants should have access to:

- A modern web browser.
- IGV Desktop or IGV-Web.
- Provided tutorial data bundle containing:

- Gene annotation tracks.
- ATAC-seq, DNase-seq and histone ChIP-seq signal and peak files.
- Ensembl Regulatory Features and activity files.
- ENCODE/SCREEN cCRE annotation files.
- R/Python notebooks.

Full day schedule

09:00 – 09:15 Opening and initial setup

Who

- Gabriela A. Merino

Topics

- Welcome and tutorial goals.
- Instructors' introductions.
- Overview of agenda and tools.
- Computing setup check (Internet access, IGV browser).
- Tutorial datasets and repositories.

09:15 – 10:00 Session 1 — Regulatory Genomics and Epigenomic Evidence

Who

- Gabriela A. Merino
- Garth Ilesley

Objectives

- Establish shared conceptual vocabulary for regulatory genomics.
- Explain the relationship between epigenomic assays and regulatory inference.
- Introduce large-scale community efforts generating epigenomic evidence useful for regulatory elements inference.
- Present different alternatives for annotating cis-regulatory elements.
- Prepare participants to interpret browser tracks in a biological context, including signal, peaks and annotations.

Topics

- Types of regulatory elements (promoters, enhancers, open chromatin, TF-binding sites) and their biological roles.
- TSS-proximal promoter evidence.
- Distal enhancer evidence.
- Epigenomic assays and signals:
 - Open chromatin from ATAC-seq or DNase-seq.
 - Histone mark interpretation:
 - H3K4me3 as promoter-associated evidence.
 - H3K27ac as active promoter/enhancer-associated evidence.
 - H3K4me1 as enhancer-associated evidence.
 - Signal tracks versus peaks tracks.
- Large scale epigenomic projects and portals (e.g. ENCODE and FAANG).
- From experimental signal to peak calls and regulatory annotations.
- Candidate annotation versus experimentally validated function.
- Strengths and limitations of regulatory inference.

Hands-on

Aims:

- Explore epigenomic tracks.
- Correlate epigenomic tracks with gene annotation.

Instructors and participants inspect together one locus:

- Gene annotation.
- ATAC-seq or DNase-seq signal and peaks.
- H3K27ac signal and peaks.
- H3K4me1 signal and peaks, where available.
- H3K4me3 and TF ChIP-seq tracks.

Participants answer:

- Which track shows a continuous signal?
- Which track contains peaks?
- Which track is gene annotation?
- Does open chromatin coincide with a promoter, an enhancer-like region, or neither?
- Is there a region with ATAC/DNase evidence but no nearby TSS or histone-mark support?

Expected Output:

- A short interpretation note distinguishing:
 - Signal evidence.
 - Peak evidence.
 - Gene annotation context.
 - Candidate regulatory interpretation.
 - Remaining uncertainty.

10:00 – 10:30 Session 2 — Part I: ENCODE: Experimental Data.

Who

- Gabriela A. Merino

Objectives

- Introduce ENCODE as a source of experimental evidence and inferred cis-regulatory elements.

Topics

- ENCODE accessions, biosamples, assays, processed files, and metadata.
- ENCODE experimental evidence versus SCREEN cCRE annotations.

Hands-on

Aims:

- Understand the ENCODE Data Portal.

Participants:

- Search for a reference epigenome in the ENCODE Data portal.
- Explore the experiments and processed files linked to the reference epigenome.

Tools

- ENCODE data portal.

10:30 – 10:45 Coffee Break

10:45-11:30 Session 2 — Part II: ENCODE: cCREs, and SCREEN.

Who

- Gabriela A. Merino

Objectives

- Understand the ENCODE registry of candidate cis-regulatory elements and the SCREEN portal.

Topics

- cCRE classes such as promoter-like signatures and enhancer-like signatures.
- Registry of cCREs and SCREEN web portal.
- SCREEN navigation:
 - Searching by gene, region, or cCRE identifier.
 - Inspecting cCRE classes.
 - Viewing biosample-specific activity.
 - Downloading cCRE files and matrices.

Hands-on

Aims:

- Explore cCREs in SCREEN.

Participants:

- Search a target locus in SCREEN.
- Inspect cCREs overlapping or near a gene or query region.
- Review cCRE class, biosample support, and linked evidence for a given reference epigenome.

Expected Output:

A short table:

Epigenome name	Experiment list	Available processed files	cCREs in locus	cCRE class for the epigenome	Supporting evidence
Example epigenome	Dnase-seq ...	Alignment file ...	cCRE IDs and agnostic annotation	cCRE class	Experimental signal in the cCRE

Tools

- ENCODE data portal.
- ENCODE SCREEN portal.

11:30 – 12:45 Session 3 — Ensembl Regulation: Regulatory Builds and Evidence

Who

- Louisse Paola Mirabueno
- Garth Ilesley

Objectives

- Introduce the Ensembl Project and the Ensembl Regulatory Build.
- Understand how Ensembl integrates epigenomic data into Regulatory Features.
- Learn how to interpret regulatory features and feature types, supporting evidence, and activity across epigenomes or cell types.
- Compare different regulatory annotation resources without promoting a single best.

Topics

- Introduction to the Ensembl Project and the Ensembl Regulatory Build.
- Ensembl Regulatory Features as standardised multi-species annotations supporting human, mouse, livestock, and aquaculture research.
- Promoter, enhancer, CTCF-binding and open chromatin feature classification.

- Supporting evidence from epigenomic experiments vs Regulatory Features.
- Cell-type, tissue, or epigenome activity.
- Changes in activity by epigenome, tissue or cell type.
- Differences between regulatory annotation frameworks:
 - Different input data collections.
 - Different annotation rules.
 - Different classification labels.
 - Different strengths for human/mouse versus multi-species use.

Hands-on

Aims:

- Introduce the Ensembl browser and the Ensembl Regulation subsite.
- Explore a genomic locus in Ensembl.
- Toggle regulatory features and evidence tracks.
- Inspect promoters vs enhancers.
- Compare different cell types.
- Compare regulatory features from different sources.

Participants use Ensembl Regulation and/or the Ensembl Genome browser to:

- Search for a reference epigenome.
- Navigate to a selected gene or regulatory locus.
- Display and explore regulatory features.
- Inspect supporting epigenomic evidence.
- Compare promoter and enhancer features.
- Compare activity across a group of epigenomes.
- Identify at least one context-dependent example.
- Compare the region with ENCODE cCREs from the previous session.
- Identify where the resources agree, differ, or provide complementary evidence.

Expected output:

- Participants write/discuss one evidence statement like: *This interval overlaps an Ensembl Regulatory feature labelled as an enhancer in the selected build, with supporting chromatin accessibility and histone-mark evidence in the inspected epigenome. The identified enhancer is also annotated as active in the studied epigenome and other epigenomes related to the same tissue.*

Tools

- Ensembl Genome Browser.
- Ensembl Regulation Subsite.

12:45 – 13:45 Lunch Break

13:45 – 14:45 Session 4 — Alternative Access to Regulatory Annotations

Who

- Gabriela A. Merino
- Louise Paola Mirabueno

Objectives

- Move from web browsers to file-based and programmatic access.
- Teach transferable data-access and interval-analysis skills.

Topics

- Accessing regulatory annotation and epigenomics data from repositories.
- Importance of versioning and release tracking.

- File formats:
 - BED for intervals.
 - bigBed for indexed intervals.
 - bigWig for continuous signal.
 - GFF3/GTF for structured annotations.
 - TSV/CSV for metadata and activity matrices.
- Coordinate conventions:
 - BED-style 0-based, half-open intervals.
 - GFF3-style 1-based closed intervals.
 - Assembly and chromosome naming harmonisation.
- Interacting with files in R and Python:
 - Parse and analyse regulatory annotations.
 - Filtering by feature type and cell type.
 - GenomicRanges workflows.
 - Integration with pandas / PyRanges.

Hands-On

Participants choose R, Python, or both.

- R path:
 - Import files with rtracklayer.
 - Represent intervals as GRanges.
 - Use GenomicRanges to filter and overlap intervals.
 - Export results as TSV.
- Python path:
 - Read annotation tables with pandas.
 - Use PyRanges for genomic interval joins.
 - Filter metadata and feature classes.
 - Export results as TSV.

Format

- Google Colab notebooks (participants can choose between R and Python)
- Guided examples with provided datasets. Example tasks:
 - Read Ensembl Regulatory Features and SCREEN cCRE subsets.
 - Read user-defined regions.
 - Inspect required columns and metadata.
 - Filter by feature class, tissue, biosample, or activity.
 - Overlap query regions with annotations.
 - Add source, version, assembly, feature ID, feature class, context, and caveat columns.

Expected Output

A provenance-aware evidence table:

query_id	resource	feature_id	feature_class	context	overlap_bp	interpretation_note
example_region	Ensembl	feature identifier	native label	tissue/cell type	overlap size	cautious evidence statement

14:45 – 15:15 Session 5 — Visualisation with Genome Browsers

Who

- Gabriela A. Merino
- Louise Paola Mirabueno

Objectives

- Use IGV to visually integrate regulatory annotations and supporting evidence.

- Learn how to arrange and customise tracks so signal, peaks, annotations, genes and user-defined regions are easily identified.
- Interpret concordance/conflicts between resources and evidence layers.

Topics

- IGV as a genome visualisation tool.
- Load/select genomes.
- Load track evidence layers:
 - User regions or variants.
 - Continuous signal tracks.
 - Peak tracks.
 - Ensembl Regulatory Features, ENCODE cCREs, etc.
 - Gene annotation tracks.
- Setting tracks:
 - Order
 - Name
 - Color
 - Display mode (collapsed, squished and expanded views)
 - Data range (for signals)
- Interpreting concordance and conflicts between annotation resources and supporting evidence tracks.

Hands-on

Participants:

- Start a clean IGV session.
- Select/load the correct reference genome.
- Load user-defined regions, signal tracks, peak tracks, regulatory elements annotation, and gene annotation.
- Rename, reorder, and color tracks by evidence layer.
- Set display options for signal and annotation tracks.
- Inspect:
 - One concordant locus.
 - One conflict between resources or evidence layers.
 - One missing-context example.
- Save an IGV session, screenshot, or launchable view.

Expected Output

A saved IGV view or screenshot plus an interpretation note: *The user interval overlaps open chromatin signal, an ATAC-seq peak, an Ensembl Regulatory feature and several ENCODE cCREs in the selected context. This supports a candidate regulatory interpretation, but the evidence is resource- and tissue-specific.*

Tools

- IGV

15:15 – 15:30 Coffee Break

15:30 – 17:15 Session 6 — Integrative Case Study: Non-coding Variant Interpretation

Who

- Gabriela A. Merino

Objectives

- Bring conceptual interpretation, resources navigation, file analysis workflows and genome browser visualisation all together
- Demonstrate a realistic analysis workflow for interpreting non-coding variants or user-defined genomic regions.
- Encourage participants to compare evidence and resources rather than seek a single definitive annotation.

Case study example

Participants analyse a disease-associated or trait-associated non-coding locus

- Example questions:
 - Are there variants in regulatory regions?
 - Do the variants or regions of interest overlap regulatory annotations?
 - Which epigenomes, cell types or tissues show relevant activity?
 - Do signal, peaks, regulatory annotations and gene context agree?
 - Which nearby genes are plausible candidates and which evidence supports this?
 - Which resource supports the inference?
 - What remains uncertain?

Workflow

- Start from user-defined variants or regions.
- Search the locus in Ensembl Regulation/IGV.
- Use provided R/Python code to overlap the query region with regulatory annotation
- Load tracks into IGV:
 - Query region/variants.
 - Epigenomic signals and peak calls on available epigenomes, cell types, or tissues.
 - Regulatory annotation from different resources
 - Gene annotation
- Interpretation and discussion.
- Write the final statement

Expected Group Output

Each group produces:

- One evidence table row or mini-table.
- One IGV screenshot or saved browser view.
- One interpretation statement.
- One follow-up analysis they would perform next.
- Example final statement: *The variant overlaps a candidate regulatory annotation in both Ensembl and SCREEN and falls within an accessibility peak in the selected epigenome. The region is near a gene with relevant biological context. This supports prioritising the variant as a candidate regulatory variant, but functional validation or additional cell-type-specific evidence would be needed before assigning causal regulatory function.*

Format

- Guided group exercise
- Discussion-driven

17:15 – 17:30 Wrap-Up

- Summary of key concepts
- When to use which resource
- Common pitfalls
- Q&A
- Links to materials and further reading

Discussion Topics

- When to use a specific regulatory annotation framework.
- When to use ENCODE, FAANG and animal genomics data sources.
- When to use browser exploration versus file-based workflows.
- How to avoid overinterpreting candidate annotations.
- How to record provenance for regulatory annotation analyses.
- How to move from candidate regulatory evidence to functional follow-up.

Take-Home Decision Rules

- Use browser resources to understand a locus and inspect evidence layers.
- Use file downloads or APIs when analysing many regions, many variants, or many samples.
- Use IGV to explore alignment between user regions, signal, peaks, annotations, and genes and integrate regulatory annotation from different sources.
- Take advantage of Ensembl Regulatory Features, ENCODE cCREs, RefSeq, FANTOM, etc. as complementary annotation systems.
- Do not treat signal, peaks, annotations, and gene links as interchangeable.
- Always report assembly, resource, version, file identifiers, tissue/cell type, and interpretation caveats.

Time	Theme Name	Session Name	Topics covered in module
09:00 - 09:15	Welcome	Welcome	Welcome Agenda overview and computers setup
09:15 - 09:30	Regulatory Genomics and Epigenomics Evidence	S1: Regulatory Genomics and Epigenomics Evidence	Type of regulatory elements Epigenomic assays From experimental data to regulatory annotation Hands-on: Epigenomic tracks and gene annotation
09:30 - 09:45			
09:45 - 10:00			
10:00-10:15	Multi-species Catalogues of Regulatory Elements	S2 – Part I: ENCODE: Experimental Data.	The ENCODE project ENCODE reference epigenomes and datasets Hands-on: The ENCODE Data Portal
10:15 - 10:30			
10:30 - 10:45	Coffee Break		
10:45 - 11:00	Multi-species Catalogues of Regulatory Elements	S2 – Part II: ENCODE: cCREs, and SCREEN	ENCODE cCREs and SCREEN cCREs cell-type annotation Hands-on: Exploring cCREs in SCREEN
11:00 - 11:15			
11:15 - 11:30			

11:30 - 11:45		S3: Ensembl Regulation: Regulatory Builds and Evidence	The Ensembl project The Ensembl Regulatory Build Regulatory Features vs supporting evidence Cell-type/epigenome activity Comparison of cis-regulatory elements resources Hands-on: Explore regulatory features and epigenomic tracks, analysis of cell-type activity and comparison of CREs annotation
11:45 - 12:00			
12:00 - 12:15			
12:15 - 12:30			
12:30 - 12:45			
12:45 - 13:00	Lunch break		
13:00 - 13:15			
13:15 - 13:30			
13:30 - 13:45			
13:45 - 14:00	Alternative Access to Regulatory Annotations	S4: From browsers to files	Accessing annotation and data files from repositories Hands-on: Parsing and analysing regulatory annotation files, filtering by feature type and cell type
14:00 - 14:15			
14:15 - 14:30			
14:30 - 14:45			
14:45 - 15:00	Visualisation with Genome Browsers	S5 - Exploring different data files on IGV	Explore epigenomics data and regulatory elements annotation on IGV Hands-on: Loading Ensembl Regulatory Feature tracks, ENCODE cCREs BED files, adding user-defined regions
15:00 - 15:15			
15:15 - 15:30	Coffee Break		
15:30-15:45	Team projects	Integrative Case Study: Non-coding Variant Interpretation	Hands-on guided activity using the presented resources and tools for analysing non-coding variants in a disease-associated locus Group presentations, ideas discussions, and future work
15:45 - 16:00			
16:00 - 16:15			

16:15 - 16:30			
16:30 - 16:45			
16:45 - 17:00			
17:00-17:15			
17:15 - 17:30	Wrap-up		

Assessment and Success Criteria

Participants will have successfully completed the tutorial if they can:

- Identify whether a track represents signal, peaks, annotations, genes, variants, or metadata.
- Explain the difference between cis-regulatory elements annotation resources like Ensembl Regulatory Features and ENCODE cCREs.
- Retrieve or load at least one regulatory annotation file.
- Run either an R or Python workflow on provided query intervals.
- Interpret an IGV view using separate evidence layers.
- Produce a cautious, provenance-aware statement about a candidate regulatory region or non-coding variant.